Neuro-Symbolic Causal Inference via Language Model Knowledge Extraction and Bayesian Networks

Divyansh Rathore

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Abstract

We build a system that extracts candidate cause–effect relations from text, assembles them into a directed acyclic graph (DAG), and uses a Bayesian network fitted to real survey data to answer interventional queries. Two extractors are compared on the same task: a rule-based baseline over spaCy dependency parses, and a fine-tuned DistilBERT relation classifier. On the held-out SemEval-2010 Task 8 test set, DistilBERT clearly outperforms the rules (F1 0.91 vs. 0.74, non-overlapping bootstrap intervals). Applied to 500 PubMed abstracts on smoking and lung disease, the picture changes. DistilBERT's precision falls to 0.45 against silver labels, its graph needs 59 edges removed to become acyclic, and it proposes arrows that reverse established facts, while the rules remain precise (0.79) but recall only 0.15. Against 14 conclusions of the US Surgeon General, the rule graph has an edge precision of 0.58 with no contradictions, and the DistilBERT graph 0.22 with two. A Bayesian network over the extracted structure, fitted to 16,904 NHANES adults with age as a confounder, answers six documented do-queries (e.g. a 9.3× causal risk ratio of smoking on COPD) in agreement with the published evidence. A single reversed arrow, *COPD* → *smoking*, which DistilBERT supported in 33 abstracts, would change that answer to 1.0 (no effect), even though both graphs fit the data equally well. Fluent text extraction is therefore not sufficient for causal reasoning; the symbolic stage is what makes the answers consistent.

1 Introduction

Scientific and news text is full of causal claims ("smoking is the leading cause of COPD"). Language models are good at recognising such statements, but a collection of extracted sentences is not a causal model. Text extraction alone cannot (i) guarantee that the extracted relations form an acyclic structure, (ii) quantify uncertainty in a calibrated way, or (iii) answer intervention queries such as "how would COPD prevalence change if nobody smoked?" consistently. Answering such queries requires a causal graph and the machinery of Bayesian networks and the do-operator [12, 13].

This project separates the two jobs in a *neuro-symbolic* pipeline (Figure 1). A **neural** stage reads text and proposes candidate (cause, effect) triples. A **symbolic** stage merges them into a DAG with evidence on every edge, removes self-loops and cycles, tests the graph against data, fits a Bayesian network, and answers interventional queries.

Contributions.

- A reproducible pipeline from text to interventional answers, combining SemEval-2010 Task 8 for training and scoring the extractors, PubMed abstracts for building the graph, and NHANES survey data for the probabilities.
- A fair comparison of a rule-based (AML) extractor and a fine-tuned DistilBERT model on identical inputs, scored both on a labelled in-domain test set and on scientific text.

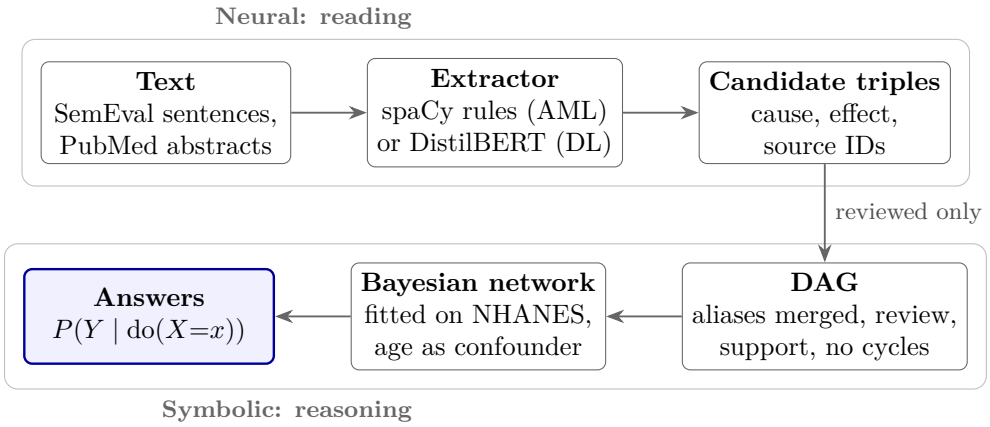


Figure 1: The neuro-symbolic pipeline. The language side proposes relations; the symbolic side enforces structure and answers interventional queries.

- A separate evaluation of the resulting graphs, by edge precision against cited Surgeon General conclusions and by the effect of the graph structure on causal answers.

2 Background

Bayesian networks and interventions. A Bayesian network is a DAG G over variables V together with conditional probability tables, factorising $P(V) = \prod_v P(v \mid \mathrm{pa}(v))$ [12]. Under a causal reading of G, the intervention $\mathrm{do}(X{=}x)$ removes all arrows into X and fixes its value (*graph surgery*). When a set Z blocks every back-door path from X to Y, the interventional distribution is identified by the adjustment formula [13]

$$P(Y \mid \mathrm{do}(X{=}x)) = \sum_z P(Y \mid X{=}x, Z{=}z)\, P(Z{=}z), \tag{1}$$

which in general differs from the observational $P(Y \mid X{=}x)$. The direction of every arrow matters: it decides which variables are adjusted for, and whether Y is affected by X at all.

Transformers for relation extraction. BERT [4] learns contextual representations by pre-training on large corpora; DistilBERT [14] is a distilled version with 40% fewer parameters that keeps most of BERT's accuracy. For relation classification, the two arguments are commonly marked with special tokens in the input [2].

Datasets considered. Corr2Cause [8], the suggested starter dataset, tests whether causal claims follow from correlation statements over abstract variables ("A correlates with B"). It contains no natural-language entities to extract, so we used SemEval-2010 Task 8 [7] for extraction, which annotates the relation between two marked nominals in natural sentences.

3 Data

Source	Role in the pipeline	Size used
SemEval-2010 Task 8 [7]	Train, tune and test both extractors	10,698 sentences (12.5% causal)
PubMed (E-utilities API)	Text the causal graph is built from	500 abstracts, 5,185 sentences
NHANES 2013–2018 + linked mortality [10, 11]	Probabilities for the Bayesian network	16,904 adults
US Surgeon General reports [16, 17]	Reference facts for graph evaluation	14 causal conclusions

Table 1: Data sources.

SemEval. Each sentence marks two entities and carries one of 19 labels (nine relation types in both directions, plus *Other*). The two *Cause-Effect* labels are positives; for *Cause-Effect(e2,e1)* the arrow is flipped so that (cause, effect) is always explicit. All other labels are negatives, which provides hard "related but not causal" examples. The official test set (2,717 sentences, 328 causal) is kept untouched until the final comparison; 15% of the official training set, stratified by label, forms the validation set (1,195 sentences). A leakage check found 19 sentences duplicated across splits, and they were removed from train/validation, leaving 6,786 training sentences (852 causal).

PubMed. We retrieved the 500 most relevant abstracts (published 2010–2024) for a query combining smoking with lung disease (COPD, emphysema, lung cancer) and cached them, so the corpus is fixed.

NHANES. Three survey cycles (2013–2018) were pooled and linked to the NCHS public-use mortality files (deaths up to 31 Dec 2019). Variables: age group (20–44 / 45–64 / 65+), smoking (never / former / current, from SMQ020 and SMQ040), self-reported doctor-diagnosed COPD (664 cases), emphysema (324), chronic bronchitis (1,003), lung cancer (54), and all-cause death during follow-up (872).

4 Methods

4.1 Task, entity schema and triples

Both extractors solve the same task: given a sentence and two marked entities, predict `e1`→`e2`, `e2`→`e1` or `none`. Every positive prediction becomes a `CausalTriple`(cause, effect, document ID, sentence ID, source, confidence). The record normalises names (lower-case, leading determiners removed) and rejects self-loops at construction time.

For PubMed we defined a small **entity schema** of 20 concepts in three groups: exposures (smoking, secondhand smoke, e-cigarettes, smoking cessation, air pollution), mechanisms (inflammation, oxidative stress, apoptosis, senescence, infection), and diseases/outcomes (COPD, emphysema, chronic bronchitis, asthma, pulmonary fibrosis, lung cancer, airflow obstruction, lung-function decline, exacerbations, mortality). Each concept lists its aliases (*smokers*, *cigarette smoke*, *CS exposure* → `smoking`). These are matched case-insensitively, with the longest match winning. An ignore list removes false friends: "cell *death*" is not mortality. Bare "fibrosis" was excluded after inspection, because it often meant airway fibrosis. Every pair of different concepts within one sentence is a candidate (6,051 pairs in 2,085 sentences) and is stored with entity markers, so both extractors label exactly the same pairs.

4.2 AML baseline: dependency rules

Rules walk the dependency path between the two entity heads (via their lowest common ancestor) and fire on a causal trigger with a grammatical pattern that also fixes the direction. There are eight patterns: active verb (*X causes Y*), relative clause (*X that leads to Y*), passive (*Y was caused by X*), participle (*Y caused by X*), "from"-verbs (*Y stems from X*), cause nouns (*X is a cause of Y*), result nouns (*Y is the result of X*) and a bare *from*. We started from 24 trigger verbs and 12 trigger nouns chosen by intuition. A trigger was then kept only if its precision on the *training* split was at least 0.5. This removed, for example, *make* (7%), *produce* (25%) and *from* (43%), which mostly signal Product-Producer and Entity-Origin relations, and kept nine triggers (*arise, cause, contribute, generate, lead, reason, result, stem, trigger*).

4.3 DL model: fine-tuned DistilBERT

We fine-tuned `distilbert-base-uncased` [14, 18] as a three-way sequence classifier over the marked sentence, adding `<e1>`, `</e1>`, `<e2>`, `</e2>` as special tokens [2]. Training used AdamW [9] (learning rate 5×10^{-5}, weight decay 0.01, batch size 32, linear decay after 10% warm-up, gradient clipping at 1.0) for four epochs. The epoch with the best validation F1 was kept. Training took about 20 minutes on an Apple M3 GPU. The softmax probability of the predicted class is stored as the triple's confidence.

4.4 From triples to a DAG

Triples from one extractor are merged into a weighted directed graph. The same fact stated twice in one sentence counts once, and every edge records its supporting abstracts and sentences. Support is counted in *abstracts*, so that one paper repeating a claim does not outweigh independent papers. Following the brief, only reviewed predictions should reach the graph: every edge is exported with an example sentence to a review file, and edges a reviewer marks *no* are removed. Edges supported by fewer than two abstracts are then dropped. Finally, while a cycle remains, its weakest edge (fewest abstracts, then fewest sentences, then a fixed order) is removed and logged.

4.5 Bayesian network

The network uses every text-DAG edge whose endpoints are both measured in NHANES (11 edges; 8 edges touching unmeasured mechanisms are left out), plus *age* $\rightarrow$ *every node*. Age is not in the entity schema, so no extracted edge mentions it, but it is a known confounder of smoking history and disease. It was added from domain knowledge and documented as such. Before fitting, the structure is tested in two ways: BIC scores [15] of alternative graphs, and chi-square tests of the conditional independences the graph implies. Conditional probability tables are estimated with a BDeu prior (equivalent sample size 10) [6], because some parent configurations are rare. Observational queries use variable elimination; interventional queries use pgmpy's causal inference engine [1], which applies the back-door adjustment (1) (here $Z = \{\text{age}\}$).

4.6 Evaluation protocol

Extraction and the graph are evaluated separately, as the brief requires.

Extraction metrics. Precision, recall and F1 on the causal class; a prediction counts as correct only if it is causal *and* has the correct direction. 95% intervals for F1 come from 1,000 bootstrap resamples [5].

PubMed silver labels. PubMed has no gold labels, so an instruction model (Claude) labelled a sample. This is permitted by the brief ("a carefully evaluated instruction model"). A written

guideline was fixed first: a pair is causal only if the sentence *claims* that one concept causes, leads to, contributes to or raises the risk of the other, and mere association is not causal. Labelling was blind to both extractors' predictions. To evaluate the labeller, it first labelled 150 SemEval test sentences (75 causal, 75 not) without seeing their human labels. The PubMed sample is stratified by the extractors' agreement (both causal / rules only / DistilBERT only / neither; 60, 20, 110 and 110 pairs). Each labelled pair carries weight $w_s = N_s/n_s$, the stratum size divided by the number sampled, so that weighted counts estimate performance over all 6,051 pairs:

$$\widehat{P} = \frac{\sum_i w_i \,[\hat{y}_i = y_i \neq \text{none}]}{\sum_i w_i \,[\hat{y}_i \neq \text{none}]}, \qquad \widehat{R} = \frac{\sum_i w_i \,[\hat{y}_i = y_i \neq \text{none}]}{\sum_i w_i \,[y_i \neq \text{none}]}. \tag{2}$$

Bootstrap resampling is done within strata.

Graph metrics. Each edge is judged by fixed rules against 14 causal conclusions drawn from the US Surgeon General's reports [16, 17]. An edge is *established* if it matches a conclusion (e.g. smoking → COPD: "the evidence is sufficient to infer that smoking is the dominant cause of COPD"). It *contradicts* if it reverses an established edge, makes mortality a cause, or makes anything cause an exposure. Otherwise it is *unverified*. Edge precision is the share of established edges.

5 Results

5.1 Extraction on SemEval

	Validation			Test (held out)	
Extractor	P	R	F1	P / R	F1 [95% CI]
Rules, all triggers	0.56	0.63	0.59	–	–
Rules, pruned triggers	**0.94**	0.49	0.65	**0.94** / 0.61	0.74 [0.69, 0.78]
DistilBERT	0.88	**0.89**	**0.88**	0.90 / **0.91**	**0.91** [0.88, 0.93]

Table 2: Directional extraction scores on SemEval. Pruning the trigger list on the training split raised the rules' validation precision from 0.56 to 0.94.

DistilBERT's validation F1 rose from 0.83 (epoch 1) to 0.88 (epoch 3) and stayed level in epoch 4, while the training loss kept falling (0.32 → 0.015), the usual onset of overfitting. On validation, DistilBERT recovered 59 causal sentences the rules missed and missed none that the rules found. Neither model predicted a wrong direction on a causal validation sentence. The rules' misses involve phrasing outside the trigger list (*turning into*), pruned ambiguous cues (*headaches from mold*) and implied causation. On the test set, the bootstrap intervals do not overlap (Table 2).

5.2 Labeller check and extraction on PubMed

Against SemEval's human labels, the labeller agreed on 96.0% of 150 sentences (Cohen's κ = 0.93 [3]), with precision 1.00 and recall 0.92. All six disagreements were borderline SemEval conventions labelled *none* (e.g. "radiation *from* the microwaves"). Of the 300 PubMed pairs, 129 were labelled causal.

	Scores				Estimated pairs (of 6,051)			
	P	R	F1	F1 95% CI	correct	wrong dir.	false alarm	missed
Rules	**0.79**	0.15	0.25	[0.20, 0.32]	223	13	**47**	1,275
DistilBERT	0.45	**0.42**	**0.43**	[0.36, 0.52]	**628**	23	744	**860**

Table 3: PubMed extraction against silver labels, weighted to all 6,051 candidate pairs. The right block estimates how many of all candidate pairs fall into each outcome.

Domain shift. DistilBERT's precision falls from 0.90 on SemEval to 0.45 on PubMed. Three diagnostics explain why. First, it calls about 24% of PubMed pairs causal (rules: 3–6%), nearly independent of how many concepts share the sentence. SemEval always marks a single pair per sentence, so the model learned to recognise causal *sentences* rather than causal *pairs*. Second, it reverses direction on unfamiliar phrasing: "The most common cause of COPD is chronic tobacco inhalation" was labelled COPD → tobacco. Third, confidence does not separate right from wrong. The median confidence of its PubMed causal calls is 0.96, and clearly wrong calls receive 0.99, so a confidence threshold would remove correct and incorrect calls alike.

5.3 The extracted graphs

From the same 6,051 pairs, the rules produced 54 distinct edges. Requiring two supporting abstracts left 20, and breaking one cycle (COPD ↔ emphysema, a 2–2 tie) left a DAG of 13 concepts and 19 edges (Figure 2a). DistilBERT produced 178 edges, and **59 edges** had to be removed to reach a DAG of 18 concepts and 59 edges. Among the edges removed were *COPD* → *smoking* (33 abstracts) and *mortality* → *smoking* (12 abstracts). The correct direction smoking → COPD survived with 37 abstracts, a margin of four.

DAG	Edges	Established	Unverified	Contradicts	Edge precision	Facts recovered
Rules	19	11	8	**0**	**0.58**	11 / 14
DistilBERT	59	13	44	2	0.22	**13 / 14**

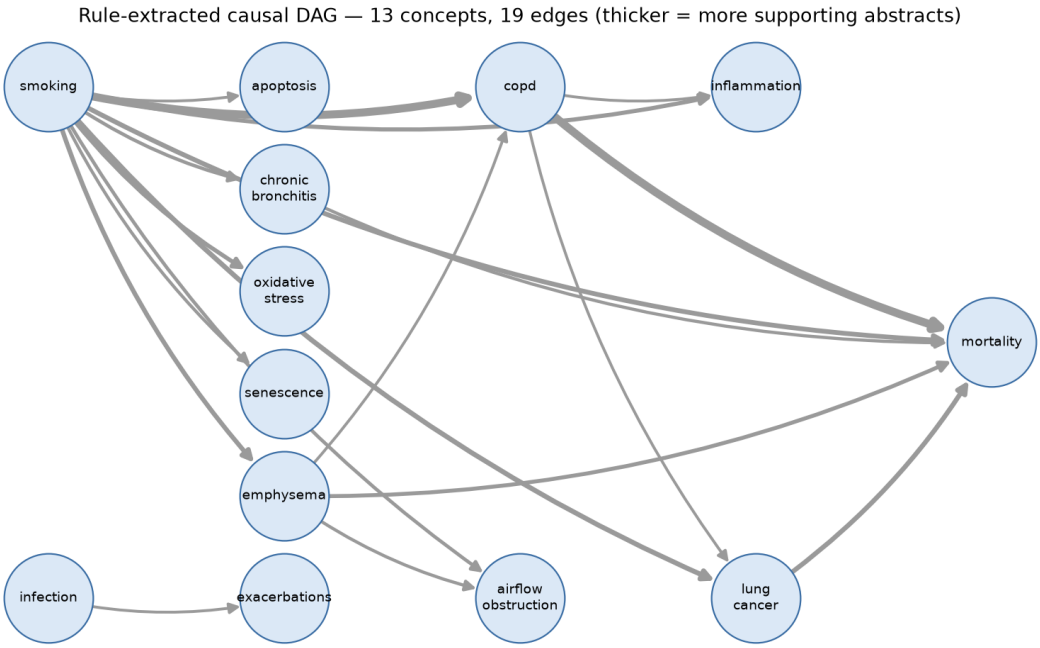
Table 4: Graph evaluation against 14 Surgeon General conclusions. DistilBERT's contradicting edges are smoking → air pollution and asthma → smoking.

5.4 Testing the structure on NHANES

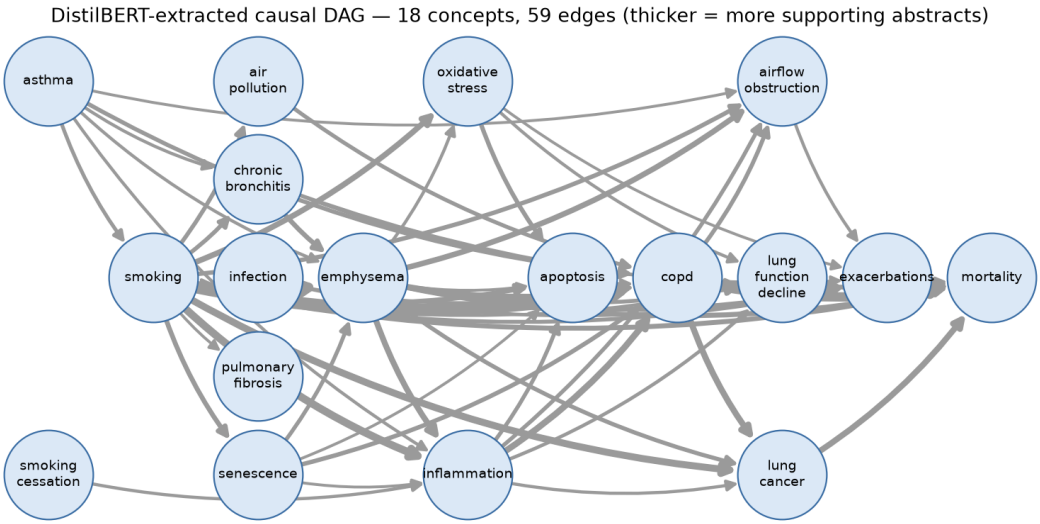
Candidate structure	BIC (higher is better)
No arrows	−46,728
Text arrows only	−45,740
Age arrows only	−45,276
Text + age (rules DAG)	−45,058
Text + age (DistilBERT DAG)	**−45,045**

Table 5: Fit of alternative structures to the NHANES data.

The extracted arrows add information beyond age (Table 5). Of the four conditional independences implied by the rules graph, two are rejected ($p < 10^{-90}$): chronic bronchitis is strongly associated with COPD and with emphysema even within the same age and smoking group. Clinically, chronic bronchitis and emphysema are the two main forms of COPD. Papers express this with *includes* rather than causal wording, so neither extractor captured it as a cause–effect pair. The DistilBERT graph fits slightly better because it contains chronic bronchitis → emphysema. We reported these rejections and did not add the edges by hand.



(a) Rules: 13 concepts, 19 edges.



(b) DistilBERT: 18 concepts, 59 edges.

Figure 2: Final DAGs extracted from the same 500 PubMed abstracts. Columns follow the causal order; thicker arrows have more supporting abstracts.

5.5 Seeing versus doing

	Never		Former		Current		Causal RR
Outcome (%)	see	do	see	do	see	do	current / never
COPD	1.02	1.12	7.16	5.24	8.86	10.42	9.3
Emphysema	0.35	0.37	3.89	2.82	4.36	5.35	14.3
Chronic bronchitis	3.57	3.73	8.02	7.25	10.61	10.87	2.9
Lung cancer	0.08	0.09	0.93	0.62	0.45	0.57	6.6
Mortality	3.48	3.85	**9.33**	**6.40**	**5.54**	**6.93**	1.8

Table 6: Observational $P(Y \mid \text{smoking})$ ("see") versus interventional $P(Y \mid \text{do(smoking)})$ ("do"), with RR the causal risk ratio. The network's "see" values match the raw survey rates to within 0.2 percentage points.

Observationally, former smokers die more often than current smokers (9.3% vs. 5.5%), which would suggest that quitting is harmful. Former smokers are much older on average. Under do(·), every group has the population's age distribution, and the order reverses (6.4% vs. 6.9%). The same adjustment lowers the apparent COPD risk of former smokers and raises that of current smokers.

5.6 Documented causal queries

	Question	Formal query	Answer
Q1	How much more COPD if everyone smoked vs. nobody?	$\frac{P(\text{copd}\mid\text{do(current)})}{P(\text{copd}\mid\text{do(never)})}$	9.3×
Q2	Share of COPD that would not exist if nobody had smoked	$\frac{P(\text{copd})-P(\text{copd}\mid\text{do(never)})}{P(\text{copd})}$	72%
Q3	COPD risk, former vs. current smokers (same age mix)	$P(\text{copd} \mid \text{do}(\cdot))$	5.2% vs. 10.4%
Q4	Effect of smoking on lung cancer	risk ratio under do	6.6×
Q5	Effect of smoking on death during follow-up	risk ratio under do	1.8×
Q6	Effect of COPD on death during follow-up	$P(\text{death} \mid \text{do(copd=yes)})$ vs. do(no)	8.2% vs. 4.8%

Table 7: Interventional queries answered by the fitted network. The directions of Q1, Q4, Q5 and Q6 agree with Surgeon General conclusions [17]; the sources state that these effects exist, not their exact size.

5.7 How much does the extracted structure matter?

Fitting the network on the DistilBERT DAG instead of the rules DAG leaves the smoking queries essentially unchanged (Q1 9.32 vs. 9.32; Q5 1.80 vs. 1.78). In both graphs, age is smoking's only parent, so the adjustment set is the same. Errors *below* a cause do not change its total effect. Errors *into* a cause do: in a network over age, smoking and COPD, reversing the single arrow to COPD → smoking turns the causal risk ratio of smoking on COPD from **9.3** into **1.0**. Under that graph, intervening on smoking cannot affect something that precedes it. The two structures are Markov equivalent and fit the data equally well, so the data cannot choose between them. DistilBERT supported the reversed arrow in 33 abstracts, and cycle breaking rejected it by a margin of four.

6 Discussion

The extractors trade precision for recall, and neither is enough alone. On in-domain text the fine-tuned model is clearly better. On scientific text, it recovers about three times as many causal pairs as the rules but adds more false alarms than correct calls. The rules are precise but blind to most phrasings. For a graph that feeds causal reasoning, precision matters more than recall: an unverified extra edge is a nuisance, while a reversed edge into a cause invalidates the answer.

The symbolic stage provides what extraction cannot. Exactly as the project brief anticipated, the extracted relations alone were inconsistent: both directions for the same pair, edges contradicting established facts, and confidence scores that did not reflect correctness. Acyclicity enforcement, evidence tracking, support thresholds and tests against data turned them into a usable structure. Interventional queries answered on that structure separate the effect of smoking from that of age, which observational percentages, and text that reports them, conflate.

7 Limitations

- PubMed labels are *silver* labels from an instruction model. It was evaluated against human labels on SemEval ($\kappa = 0.93$), but no human checked the PubMed sample, and the edge review files were not completed by a human reviewer.
- The reference list holds 14 conclusions, so "unverified" edges (e.g. mechanistic links such as apoptosis) may still be true.
- NHANES diagnoses are self-reported, mortality follow-up is short (about 1–7 years), and lung cancer is rare (54 cases), so Q4 is uncertain. Age is the only confounder modelled; income, occupational exposure and pollution are not.
- Cycle breaking keeps the better-supported direction, which can fail when the wrong direction is stated more often.

8 Conclusion and future work

Fine-tuned transformers read causal language well, but on their own they do not produce a causal model that can be trusted. Combining them with a symbolic stage that enforces structure, keeps evidence, tests against data and answers do-queries yields answers that agree with the published evidence and avoid the confounding that misleads raw observation. Future work: (1) fine-tune on labelled PubMed pairs, split by abstract to avoid leakage, to address domain shift; (2) train with several marked pairs per sentence, so the model learns *which* pair is causal; (3) complete human review of the extracted edges and of a sample of the silver labels; and (4) add confounders and spirometry-based airflow obstruction (NHANES 2007–2012) to the network.

Reproducibility. All code is in one notebook (`causal_nlp.ipynb`) with a fixed seed. Every download is cached, and package versions are pinned in a full lock file. The notebook was re-run from scratch in a freshly built environment and reproduced every reported number.

References

[1] A. Ankan and A. Panda. pgmpy: Probabilistic graphical models using Python. In *Proceedings of the 14th Python in Science Conference (SciPy)*, 2015.

[2] L. Baldini Soares, N. FitzGerald, J. Ling, and T. Kwiatkowski. Matching the blanks: Distributional similarity for relation learning. In *Proceedings of ACL*, 2019.

[3] J. Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960.

[4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019.

[5] B. Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.

[6] D. Heckerman, D. Geiger, and D. M. Chickering. Learning Bayesian networks: The combination of knowledge and statistical data. *Machine Learning*, 20:197–243, 1995.

[7] I. Hendrickx, S. N. Kim, Z. Kozareva, P. Nakov, D. Ó Séaghdha, S. Padó, M. Pennacchiotti, L. Romano, and S. Szpakowicz. SemEval-2010 task 8: Multi-way classification of semantic relations between pairs of nominals. In *Proceedings of the 5th International Workshop on Semantic Evaluation*, pages 33–38, 2010.

[8] Z. Jin, J. Liu, Z. Lyu, S. Poff, M. Sachan, R. Mihalcea, M. Diab, and B. Schölkopf. Can large language models infer causation from correlation? In *International Conference on Learning Representations (ICLR)*, 2024.

[9] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[10] National Center for Health Statistics. Public-use linked mortality files (NHANES, mortality follow-up through 2019). https://ftp.cdc.gov/pub/Health_Statistics/NCHS/datalinkage/linked_mortality/.

[11] Centers for Disease Control and Prevention, National Center for Health Statistics. National Health and Nutrition Examination Survey data, cycles 2013–2014, 2015–2016, 2017–2018. https://wwwn.cdc.gov/nchs/nhanes/.

[12] J. Pearl. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann, 1988.

[13] J. Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition, 2009.

[14] V. Sanh, L. Debut, J. Chaumond, and T. Wolf. DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.

[15] G. Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978.

[16] U.S. Department of Health and Human Services. *How Tobacco Smoke Causes Disease: The Biology and Behavioral Basis for Smoking-Attributable Disease. A Report of the Surgeon General*. Atlanta, GA, 2010. https://www.ncbi.nlm.nih.gov/books/NBK53021/.

[17] U.S. Department of Health and Human Services. *The Health Consequences of Smoking—50 Years of Progress. A Report of the Surgeon General*. Atlanta, GA, 2014. https://www.ncbi.nlm.nih.gov/books/NBK294322/.

[18] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, et al. Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP: System Demonstrations*, pages 38–45, 2020.